

Leatt ZA Ecommerce BI and ML Ebook

Catalog extraction, synthetic ecommerce data, KPI modelling, ML segmentation, forecasting, and Microsoft Fabric implementation plan.

Metric	Value
Transaction lines	2,000,000
Catalog variants extracted	11,354
Net revenue	R3,994,247,413
Gross margin rate	48.5%
Return value rate	4.9%
Repurchase model AUC	0.810

Executive Insights

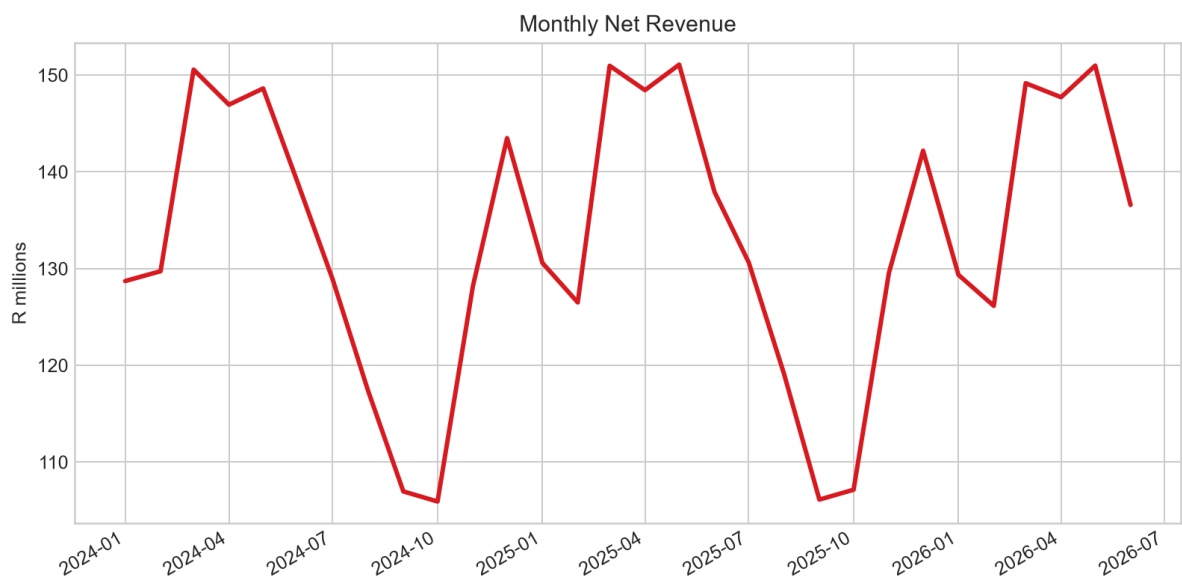
Apparel is the largest simulated revenue category at R1,759,430,045. It should anchor homepage merchandising, paid search, and bundle strategy.

The strongest product variant by revenue is Jacket ADV FlowTour 5.5 - Women at R5,714,161. Use it as a hero SKU and cross-sell relevant accessories.

Gauteng is the top market at R1,236,840,522. Regional budget allocation should follow demand density while protecting margin.

Revenue changed 6.1% from 2024-01 to 2026-06 in the generated trading history, with peak season uplift visible in November and December.

The ML segmentation separates customers into VIP, Retain, Grow, and Nurture groups. VIP and Retain audiences should receive early access, replacement-cycle campaigns, and product education.



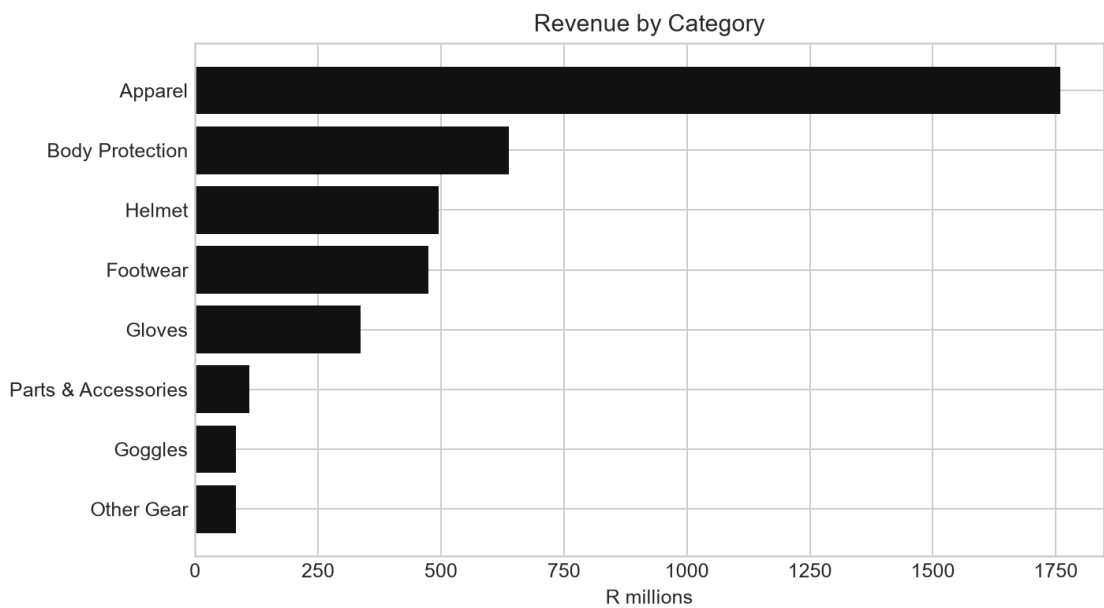
Catalog Source

Public product and variant data was extracted from Leatt ZA's Shopify-style product JSON endpoint, then normalized into a product variant dimension.

About Us – Leatt ZA Skip to content FREE SHIPPING FOR ALL ORDERS OVER R1000. Ts&Cs; Apply Search MOTO Toggle menu Apparel Toggle menu All Apparel Gloves Jackets & Vests Jerseys Pants Ride Kits Socks & Sleeves Boots Toggle menu Boots Parts & Accessories Goggles Toggle menu Goggles Lenses Parts & Accessories Helmets Toggle menu Helmets Parts & Accessories Hydration & Bags Toggle menu Hydration & Bags Parts & Accessories Neck Braces Toggle menu Neck Braces Parts & Accessories Protection Toggle menu All Protection Body Protection Chest Protection Elbow Guards Impact Shorts Knee Braces Knee Guards Neck Braces Wrist Braces Parts & Accessories Rider Collection Toggle menu Men Women Youth MTB Toggle menu Apparel Toggle menu All Apparel Bibs & Liners Gloves Jackets & Vests Jerseys Mono Suits Pants Ride Kits Shorts Socks & Sleeves Components Toggle menu All Components Grips Handlebars Pedals Stems Parts & Accessories Eyewear Toggle menu Goggles Sunglasses Lenses Parts & Accessories Helmets Toggle menu Helmets Parts & Accessories Hydration & Bags Toggle menu Hydration & Bags Parts & Accessories Neck Braces Toggle menu Neck Braces Parts & Accessories Protection Toggle menu All Protection Body Protection Chest Protection Elbow Guards Impact Pants Knee Braces Knee Guards Neck Braces Wrist Brace

Category	Variants	Avg price
Apparel	4,723	R1,899
Footwear	1,581	R3,146
Body Protection	1,511	R2,667
Gloves	1,235	R945
Helmet	1,206	R2,931
Parts & Accessories	412	R1,455
Goggles	316	R1,239
Other Gear	305	R3,135
Hydration	53	R1,570
Gift Card	7	R1,929

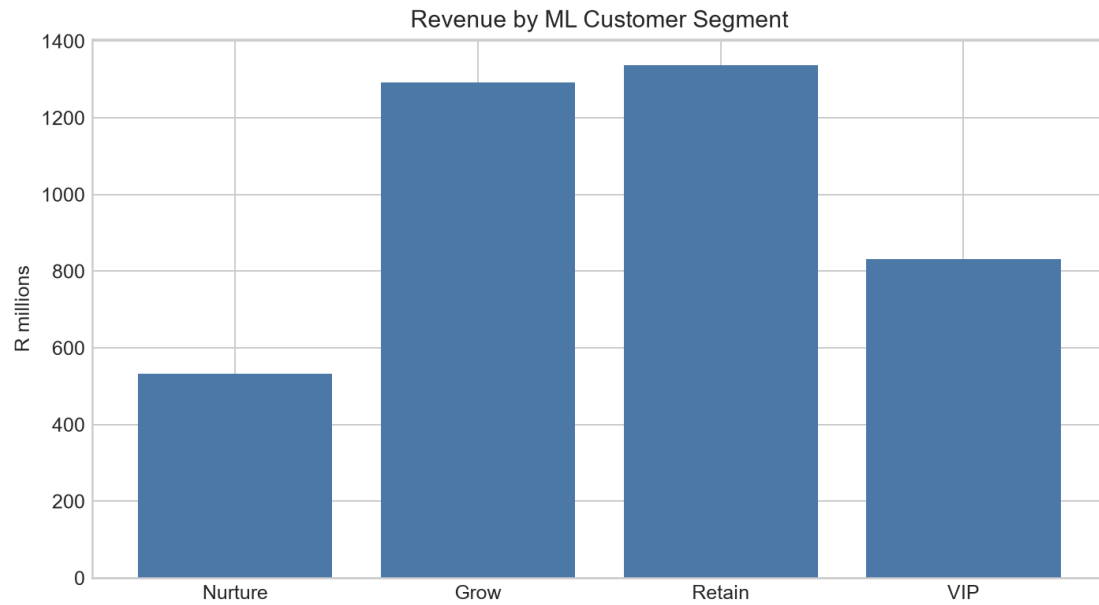
Category and Product Performance



Product	Category	Revenue	Margin
Jacket ADV FlowTour 5.5 - Women	Apparel	R5,714,161	R2,972,033
Pant ADV FlowTour 5.5	Apparel	R5,145,016	R2,684,621
Neck Brace 3.5	Body Protection	R4,974,136	R2,270,201
Jacket ADV FlowTour 5.5	Apparel	R4,513,374	R2,346,390
Helmet Kit ADV 7.5 Polymer with 2.5 Goggle	Helmet	R4,334,599	R1,841,144
Boots 5.5 FlexLock	Footwear	R4,216,577	R1,649,699
Pants Moto 5.5 Enduro	Apparel	R4,177,628	R2,177,298
Gloves ADV X-Flow 5.5 Short	Gloves	R4,173,380	R2,261,226
Jacket ADV Rally 5.5	Apparel	R3,960,176	R2,068,157
Helmet MTB Endurance 4.0	Helmet	R3,885,850	R1,653,086

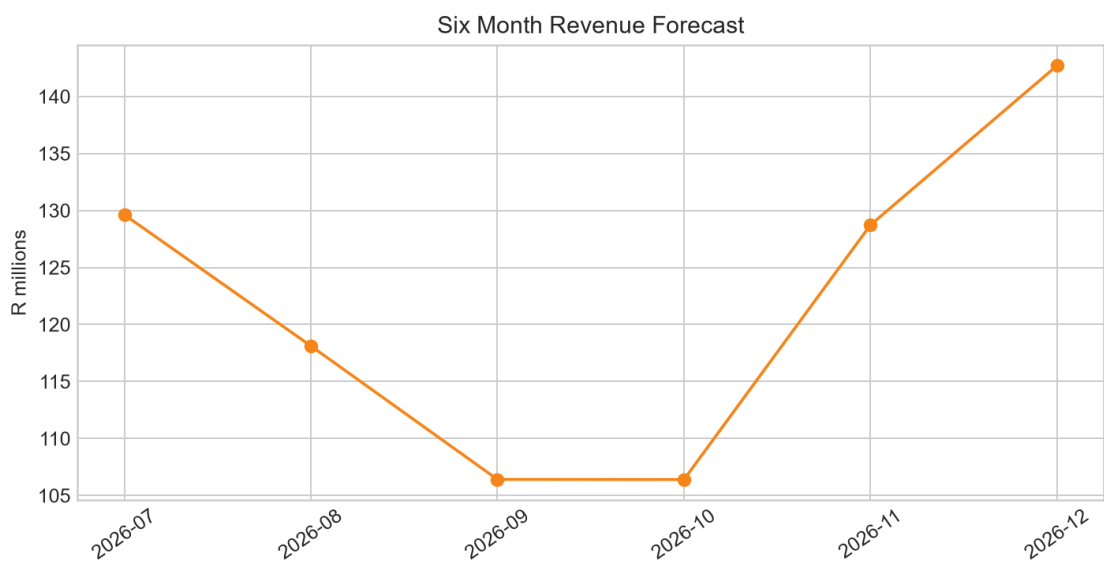
ML Customer Segmentation

The repurchase propensity model was trained on 140,000 customer examples using recency, frequency, monetary value, margin, returns, geography, rider type, loyalty tier, and acquisition channel. Test AUC: 0.810.



Segment	Customers	Revenue	Avg propensity
Nurture	79,635	R532,092,427	29.9%
Grow	48,467	R1,292,770,693	62.4%
Retain	23,559	R1,338,192,166	90.0%
VIP	7,518	R831,192,126	99.2%

Forecast and Action Plan



Month	Forecast revenue
2026-07	R129,627,593
2026-08	R118,118,723
2026-09	R106,413,145
2026-10	R106,399,810
2026-11	R128,729,963
2026-12	R142,703,616

- Prioritize high-margin protection, helmet, and accessory bundles by rider type.
- Build abandon-cart and replenishment journeys around accessories, gloves, goggles, and hydration add-ons.
- Use VIP and Retain ML segments for early product drops, service reminders, and premium support.
- Monitor return rate by category and province to find sizing, delivery, or expectation gaps.
- Publish a Fabric semantic model so Power BI can serve executive, category, customer, operations, and ML monitoring pages.

Microsoft Fabric Blueprint

Fabric area	Recommended use
Data Factory	Ingest Leatt product JSON, ad data, GA4 exports, payment/order extracts, and support data.
Lakehouse / OneLake	Store bronze raw files, silver Delta tables, and gold star schema tables.
Data Warehouse	Serve governed SQL models for finance and operations users.
Data Science	Train repurchase propensity, demand forecast, next-best-category, and return-risk models.
Power BI	Publish executive KPI, category, customer, operations, forecast, and ML monitoring report pages.
Capacity Metrics	Track CU consumption and decide when to pause, resume, or scale Fabric capacity.

Evidence and Source Addendum

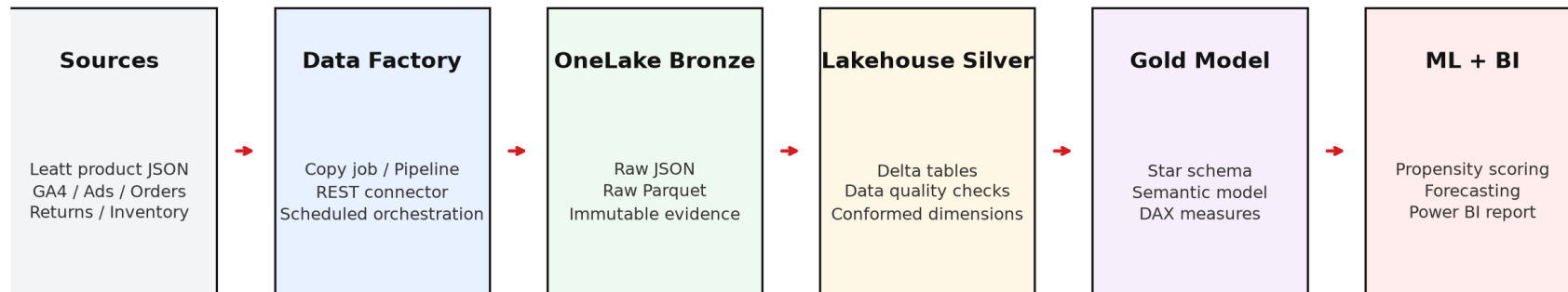
Consolidated source register, Azure/Fabric process proof, and real project screenshots for portfolio presentation.

Source	Type	Used for
Leatt ZA product JSON	Public ecommerce catalog endpoint	Product, variant, price, SKU, availability, tags, category classification seed.
Leatt ZA About Us	Public website page	Brand and business context in the ebook narrative.
Leatt ZA Moto Boots collection	Public website page and user screenshot	Visual proof of product assortment/pricing and portfolio evidence.
Microsoft Fabric Data Factory documentation	Official Microsoft documentation	Fabric ingestion, transformation, and orchestration process design.
Azure Data Factory documentation	Official Microsoft documentation	Azure ETL/data integration positioning and ADF process evidence.
Azure Data Factory Copy Activity overview	Official Microsoft documentation	Copy activity pattern for source-to-cloud movement.
Microsoft Fabric OneLake overview	Official Microsoft documentation	Unified data lake and OneLake architecture explanation.
Microsoft Fabric Lakehouse overview	Official Microsoft documentation	Lakehouse, Delta, Spark/SQL, Power BI integration design.
Microsoft Fabric subscription / Azure F SKO	Official Microsoft documentation	Confirming Fabric capacity can be purchased through Azure portal.
Power BI Direct Lake overview	Official Microsoft documentation	Power BI semantic model design for high-volume Delta tables in OneLake.
Generated synthetic ecommerce transactions	Generated dataset	Million-row transaction fact table for BI/ML modeling; synthetic, not Leatt actual sales.
Generated SQLite warehouse	Generated local warehouse	Dimensional model, fact table, and BI aggregations.

Azure and Fabric Process Proof

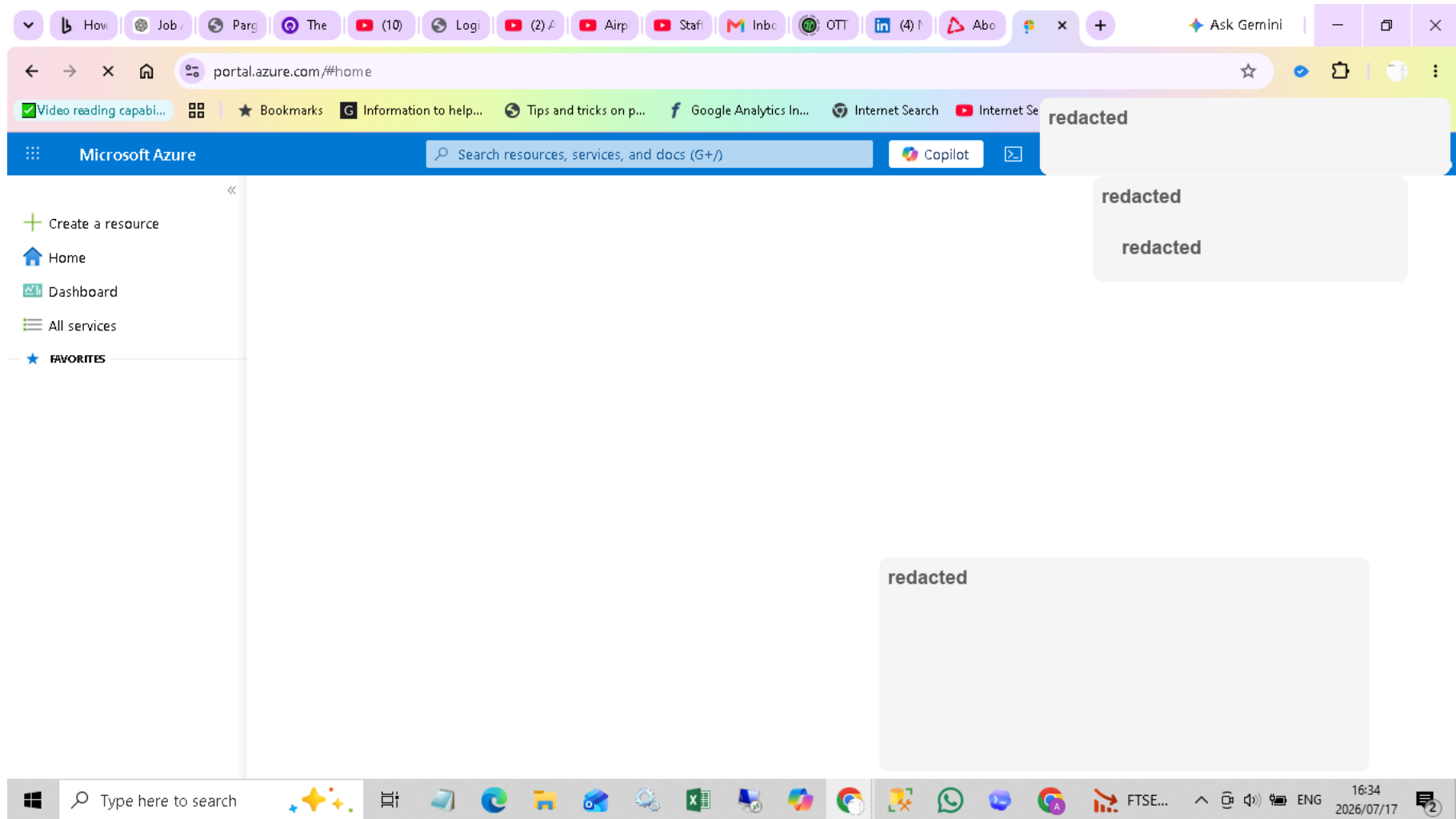
The following blueprint shows the intended Azure/Fabric implementation path: Data Factory ingestion, OneLake bronze storage, Lakehouse Delta modeling, Power BI semantic layer, and ML scoring/forecasting.

Azure / Microsoft Fabric Ecommerce BI and ML Process



Designed for Fabric Data Factory ingestion, OneLake/Lakehouse storage, Direct Lake/Power BI reporting, and Fabric Data Science models.

Azure portal opened in the user's signed-in browser. Used as environment proof for Azure/Fabric work context.



Evidence type: user-provided live screenshot. Azure private details redacted where applicable.

Azure portal home showing active subscription context. Account and cost details redacted for privacy.

The screenshot displays the Azure portal home page. The browser's address bar shows `portal.azure.com/#home`. The top navigation bar includes the Microsoft Azure logo, an 'Upgrade' button, a search bar, and a 'Copilot' button. The left sidebar lists navigation options: 'Create a resource', 'Home', 'Dashboard', 'All services', 'FAVORITES', 'Application Templates', 'All resources', 'Resource groups', 'App Services', 'Function App', 'Azure SQL Database', 'Azure Cosmos DB', 'Virtual machines', 'Load balancer', 'Storage accounts', 'Virtual networks', and 'Microsoft Entra ID'. The main content area features a 'Welcome back, Anthony' message. Below this, there are four action tiles: 'Start with a template', 'Create a resource', 'Build an AI agent', and 'Import code from GitHub'. At the bottom, there are two summary tiles: 'Alerts and service issues' (showing 'No alerts set up') and 'Costs' (showing 'redacted'). The Windows taskbar at the bottom shows the search bar and various application icons. The system clock indicates 16:35 on 2026/07/17.

Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot

redacted

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Welcome back, Anthony

Start with a template
Deploy in minutes using pre-made templates.

Create a resource
Choose a service to create a resource in your subscription.

Build an AI agent
Create and manage AI apps and agents using the latest models.

Import code from GitHub
Connect your GitHub account and deploy existing repositories.

Alerts and service issues Last updated: a few seconds ago

No alerts set up
Detect and respond to issues early, like errors or performance issues

Service issues
0
Active issues affecting Azure services reported by Microsoft

Costs Last updated: a few seconds ago

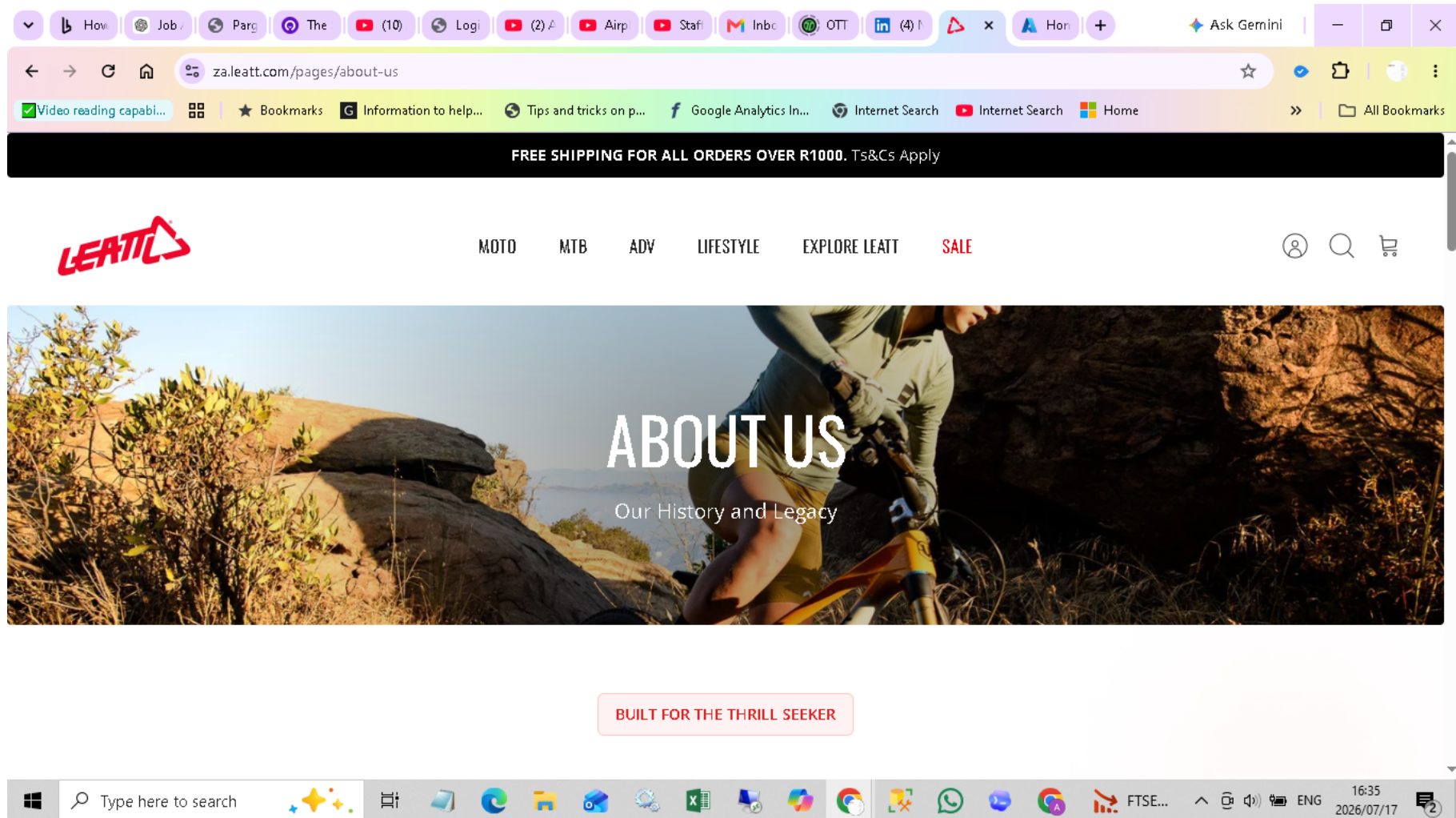
redacted

Type here to search

16:35
2026/07/17

Evidence type: user-provided live screenshot. Azure private details redacted where applicable.

Leatt ZA About Us page used for business context and brand narrative.



Evidence type: user-provided live screenshot. Azure private details redacted where applicable.

Leatt ZA Moto Boots collection page showing product/pricing context from the ecommerce source.

The screenshot shows the Leatt ZA Moto Boots collection page. The browser address bar displays the URL za.leatt.com/collections/moto-boots. The page features a navigation bar with the Leatt logo and categories: MOTO, MTB, ADV, LIFESTYLE, EXPLORE LEATT, and SALE. Below the navigation bar, four motorcycle boots are displayed in a grid:

- Boots 6.5**: R 10,799.00 ZAR
- Boots 5.5 FlexLock**: From R 8,799.00 ZAR
- Boots 5.5 FlexLock Enduro**: From R 8,799.00 ZAR
- Boots 4.5 HydraDri**: From R 7,499.00 ZAR

The Windows taskbar at the bottom shows the search bar with the text "Type here to search" and various application icons. The system clock indicates the time is 16:37 on 2026/07/17.

Evidence type: user-provided live screenshot. Azure private details redacted where applicable.